

Residual Adaptive Blending Units for Physics informed Neuronal Networks

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Maestría en Matemáticas Aplicadas

Convection Equation

Se introduce *Adaptative Blending Units*:

$$f(x) = \sum_{i=1}^N G(\alpha_i) \sigma_i(\beta_i x).$$

Se considera la ecuación de convección en lineal en 1D:

$$\begin{aligned} u_t + \beta u_x &= 0, \quad x \in [0, 2\pi], \quad t \in [0, 1], \\ u(0, x) &= \sin(x), \\ u(t, 0) &= u(t, 2\pi). \end{aligned}$$

Donde $\beta = 64$. Usaron puntos de colocación fijos de $N_r = 6400$, $N_{ic} = 512$ y $N_{bc} = 200$. La arquitectura es una fully-connected de 6 capas con 64 neuronas cada. Se dejo correr Adam por 100,000 epocas.

En la replicación se usó la misma arquitectura, los mismos puntos de colocación, el mismo optimizador. Se usaron funciones de activación $\sigma = \{\sin, \cos, \text{racional}, \tanh\}$. Se incluye algoritmos LRA, remuestreo estocástico en cada época y tasa de aprendizaje con decaimiento de coseno. Se corrió por 40,000 épocas con Adam como aproximación.

for $n = 1, \dots, S$ **do**

(a) Compute $\hat{\lambda}_i$ by

$$\hat{\lambda}_i = \frac{\max_{\theta} \{|\nabla_{\theta} \mathcal{L}_r(\theta_n)|\}}{\overline{|\nabla_{\theta} \mathcal{L}_i(\theta_n)|}}, \quad i = 1, \dots, M, \quad (40)$$

where $\overline{|\nabla_{\theta} \mathcal{L}_i(\theta_n)|}$ denotes the mean of $|\nabla_{\theta} \mathcal{L}_i(\theta_n)|$ with respect to parameters θ .
 (b) Update the weights λ_i using a moving average of the form

$$\lambda_i = (1 - \alpha)\lambda_i + \alpha\hat{\lambda}_i, \quad i = 1, \dots, M. \quad (41)$$

(c) Update the parameters θ via gradient descent

$$\theta_{n+1} = \theta_n - \eta \nabla_{\theta} \mathcal{L}_r(\theta_n) - \eta \sum_{i=1}^M \lambda_i \nabla_{\theta} \mathcal{L}_i(\theta_n) \quad (42)$$

end

The recommended hyper-parameter values are: $\eta = 10^{-3}$ and $\alpha = 0.9$.

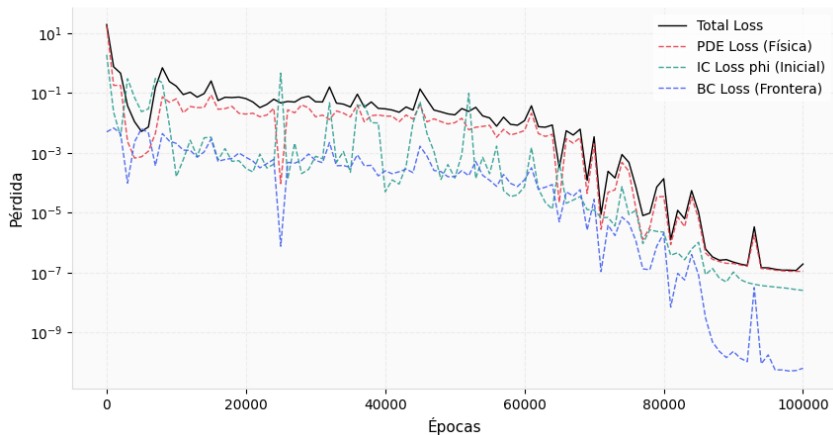
Resultados publicados

Table 5: Comparisons of standard activation functions, adaptive activation functions and their counterparts with adaptive slopes (AS) on 1D time-dependent PDEs. We report the L_2 relative error (%) for each problem and the average error rate over all problems. The better results are **bold-faced**.

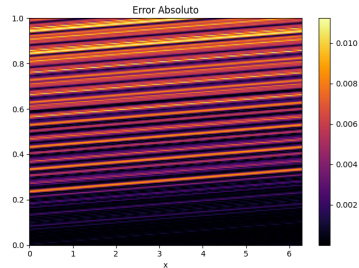
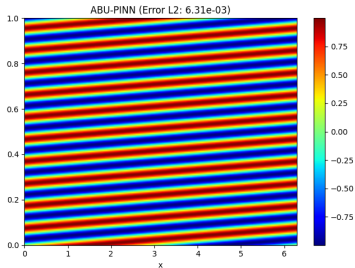
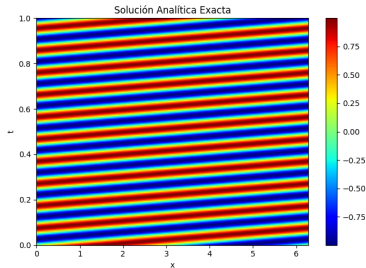
Method	Convection equation	Burgers' equation	Allen-Cahn equation	KdV equation	Cahn-Hilliard equation	Average error
sin	0.36±0.15	5.18±3.73	3.57±0.64	0.63±0.17	1.72±0.61	2.29
tanh	6.83±4.79	0.26±0.13	1.34±0.54	1.32±1.12	4.02±4.56	2.75
sigmoid	70.38±2.98	1.24±1.05	1.63±0.13	2.34±0.53	3.12±2.72	15.74
GELU	39.29±35.51	4.43±2.87	3.93±0.78	1.21±0.41	1.01±1.27	9.97
Swish	5.59±2.18	8.27±4.35	5.56±1.28	1.73±0.10	2.22±2.60	4.67
Softplus	55.39±2.46	17.75±8.11	17.72±6.04	6.23±0.44	9.67±2.57	21.35
ELU	6.67±0.96	46.34±2.36	52.55±2.95	78.95±2.57	90.77±2.07	55.66
SLAF	0.36±0.18	43.93±0.66	33.93±9.97	25.23±1.28	52.57±27.93	31.20
PAU	45.78±35.47	48.31±8.21	43.83±15.18	68.11±13.49	115.59±3.79	64.32
ACON	3.55±1.66	1.18±1.55	3.88±1.82	1.52±0.35	2.46±1.96	2.52
ABU-PINN	0.06±0.03	0.21±0.11	0.76±0.26	0.34±0.05	0.50±0.15	0.37
With AS						
sin	0.28±0.08	1.82±1.58	3.57±0.46	0.57±0.19	1.73±0.73	1.59
tanh	3.06±1.65	0.16±0.09	0.81±0.21	1.71±1.53	2.11±0.49	1.57
GELU	38.00±39.53	0.71±0.38	1.99±0.63	0.79±0.17	0.96±0.57	8.49
ABU-PINN	0.05±0.02	0.15±0.10	0.58±0.15	0.34±0.08	0.35±0.07	0.29

Función de costo

Historial de Entrenamiento



Representabilidad y Error



Cahn-Hilliard Equation

Se considera la ecuación de Cahn-Hilliard:

$$u_t - \nabla^2 (-\lambda_1 \lambda_2 h + \lambda_2 (u^3 - u)) = 0, \quad h = \nabla^2 u, \quad x \in [-1, 1], \quad t \in [0, 1], \quad (1)$$

$$u(0, x) = \cos(\pi x) - \exp(-4(\pi x)^2), \quad (2)$$

$$u(t, -1) = u(t, 1), \quad u_x(t, -1) = u_x(t, 1), \quad (3)$$

$$h(t, -1) = h(t, 1), \quad h_x(t, -1) = h_x(t, 1), \quad (4)$$

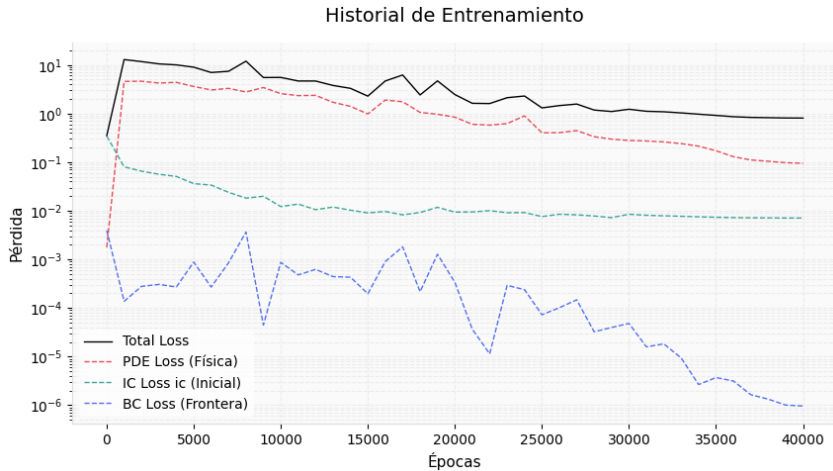
Dónde $\lambda_1 = 0,02$ y $\lambda_2 = 1$. Usaron puntos de colocación fijos de $N_r = 15000$, $N_{ic} = 256$ y $N_{bc} = 100$. La arquitectura es una fully-connected de 4 capas con 32 neuronas cada. Se dejó correr Adam por 100,000 épocas y 15,000 épocas con L-BFGS. Se usaron los mismos algoritmos que en ecuación de convección.

Resultados publicados

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Representabilidad y Error

